ELSEVIER

Signal Processing 55 (1996) 179–189

SIGNAL PROCESSING

SVD based technique for noise reduction in electroencephalographic signals

P.K. Sadasivan, D. Narayana Dutt*

Department of Electrical Communication Engineering, Indian Institute of Science, Bangalore 560012, India

Received 1 September 1994; revised 4 October 1995

Abstract

Noise reduction is a subject of importance in many applications of signal processing such as communication, instrumentation, etc. In a recent paper (Callaerts et al., 1988), singular value decomposition (SVD) has been shown to be a very robust and numerically reliable tool for extracting the desired signals from noisy data. Along similar lines, in this paper, we explore the application of SVD for extracting the actual electroencephalographic (EEG) signals from electrooculogram (EOG) contaminated EEG signals. The key point which made the SVD technique suitable for this EEG enhancement problem is the orthogonality between the EEG and EOG signals. The problem has been set up in such a way that the SVD technique could be applied directly to the recorded signals. Studies were performed on simulated as well as real signals. Signal to noise ratio (SNR) and linear prediction (LP) spectra (along with time plots) are used as measures for comparing the performance of the proposed algorithm for the cases of one EOG channel and two EOG channels. Studies on simulated as well as actual signals show that the technique is very effective in minimizing EOG artefacts from noisy EEG signals.

Zusammenfassung

Geräuschreduktion ist für viele Anwendungen der Signalverarbeitung, wie Übertragung, Meßtechnik usw. von Bedeutung. In einem kürzlich erschienenen Artikel (Callaerts et al., 1988) wurde gezeigt, daß die Singularwertzerlegung (SVD) eine sehr robuste und numerisch zuverlässige Methode ist, um gewünschte Daten aus verrauschten abzuleiten. In ähnlicher Weise untersuchen wir in dieser Arbeit die Anwendung der SVD, um die tatsächlichen elektroenzephalographischen (EEG) Signale aus Elektrookulogramme (EOG) enthaltenden EEG-Signalen abzuleiten. Die Grundlage, die die SVD-Methode für diese EEG-Verbesserung anwendbar macht, liegt in der Orthogonalität zwischen den EEG- und den EOG-Signalen, Das Problem wurde so formuliert, daß die SVD-Methode direkt auf die aufgezeichneten Signale angewendet werden kann. Sowohl simulierte als auch echte Signale wurden untersucht. Das Signalzurauschverhältnis (SNR) und die aus linearer Prädiktion hergeleiteten Spektren (neben den Zeitsignalen) wurden als Maße zum Vergleich der Leistungsfähigkeit des vorgeschlagenen Algorithmus für die Fälle eines oder zweier EOG-Kanäle verwendet. Untersuchungen bei simulierten und echten Signalen zeigen, daß die Methode sehr effizient ist, um EOG-Artefakte in verrauschten EEG-Signalen zu minimieren.

* Corresponding author. Tel.: 91-080-3092742; fax: 91-080-3341683; e-mail: dndutt@ece.iisc.ernet.in.

0165-1684/96/$15.00 © 1996 Published by Elsevier Science B.V. All rights reserved
PII S0165-1684(96)00129-6

Résumé

La réduction de bruit est un sujet important dans beaucoup d'applications de traitement du signal, comme les communications, l'instrumentation, etc. Dans un article récent (Callaerts et al., 1988), il a été montré que la décomposition en valeurs singulières (DVS) est un outil très robuste et numériquement fiable pour extraire les signaux désirés de données bruitées. De manière similaire, cet article explore l'application de la DVS à l'extraction des signaux électroencéphalographiques (EEG) à partir de signaux d'EEG contaminés par l'électrooculogramme (EOG). Le point clé qui rend la technique de SVD utilisable dans le problème du rehaussement de l'EEG est l'orthogonalité entre les signaux EEG et EOG. Le problème a été posé de telle manière que la DVS puisse être appliquée directement aux signaux enregistrés. Des études ont été faites sur des signaux simulés aussi bien qu'expérimentaux. Les spectres du rapport signal-sur-bruit et de la prédiction linéaire (de même que les graphes temporels) sont utilisés comme mesures de comparaison des performances de l'algorithme proposé dans le cas d'un canal EOG et de deux canaux EEG. L'expérience sur des signaux réels et expérimentaux montre que la technique est trés efficace pour minimiser les artéfacts d'EOG dans les signaux d'EEG bruités.

Keywords: Singular value decomposition; Electroencephalogram; EOG artefacts; Noise reduction

1. Introduction

A major challenge in many signal processing applications is the problem of signal contamination or artefacts. Hence, noise reduction is a subject of wide spread interest in communications, physical systems and experimental measurements. Several methods have been suggested in the literature for minimizing these artefacts. The technique of singular value decomposition (SVD), which is very robust and numerically reliable, plays a key role in extracting the actual signals from the recorded data matrix. This property of the SVD based technique has been exploited effectively in this paper to extract actual EEG signals from EOG contaminated EEG signals.

The brain generates electrical activity and this can be recorded from the scalp by placing electrodes on the intact skull. This recorded representation of the brain's of electrical potentials is called electroencephalogram (EEG). EEG is one of the commonly used noninvasive technique for understanding the brain functions, abnormalities, injuries, etc. Compared to the manual processing of EEG signals, automatic analysis by computers (e.g. signal processing tools) is desirable in both clinical and experimental EEG work for the following reasons: to obtain precise characterization and quantification, to avoid errors due to subjectivity, to minimize processing time, etc. However, automated analysis of the EEG is bedeviled by the lack of a reliable means for removing artefacts and, in particular, a means for distinguishing between artefacts and pathological waves of cerebral origin.

Artefacts which are almost always present, are caused by sources both external and internal to the body. Internal sources include head and body movements, muscle and cardiac activities, as well as eye movements, etc. Common external sources are mains interference and electrode movements due to poor contacts with the skin. Artefacts reduce the clinical usefulness of biomedical signals and make both manual and automatic analysis difficult, and, in some cases, impossible because of the similarity between artefacts and the pathological signals of interest. For example, in electroencephalography, the artefacts which result from eye movements is a constant source of difficulty in distinguishing normal activities from abnormal ones.

Normal EEG signals are often of very small amplitude, of the order of few microvolts [6]. Eye movements produce large electrical potentials, called electrooculogram (EOG) or ocular artefacts, in the millivolt range and these activities spread across the head or scalp to contaminate the EEG signals. The presence of EOG signals severely affects the performance of the EEG processing systems, the aim of which is to extract valid information from the recorded signals and present it in a more convenient form for interpretation. Effective elimination of artefacts from the collected data is an

essential step in preparing the data for analysis. This necessitates the removal of EOG signals from contaminated EEG signals and this forms an important part of the automation of EEG signals.

Studies have been reported in various publications for minimizing EOG artefacts from contaminated EEG signals. Depending upon the principle of operation, these techniques can be broadly classified into three categories, viz. (i) rejection techniques, (ii) eye fixation techniques and (iii) EOG subtraction techniques. The first one is the simplest in which the EEG tracings which contain sufficient EOG artefacts are discarded from the analysis [1]. This method can lead to the loss of unacceptable amount of data and sometimes it makes the data even unrepresentative [5, 15]. In the eye fixation method, the subjects are asked to fix their eyes on a target and sometimes to avoid moving their eyes at critical times during the experiment [11]. The most popular way of minimizing EOG artefacts from EEG signals falls in the third category where the EOG subtraction techniques have been used. Artefacts minimization using computers is based on different techniques under this category. Extensive review of all these techniques are covered in [5, 2, 4, 7, 13].

In this paper, we show how singular value decomposition can be effectively utilized for extracting the pure EEG from EOG contaminanted EEG signals. This is made possible by the fact that EEG and EOG signals are uncorrelated with each other [7]. We show below how the EOG artefact minimization problem can be modeled so that the SVD technique described in Section 2 can be applied for extracting artefact free EEG signals from contaminated EEG signals.

This paper is organised in the following way. Section 2 introduces the SVD method of signal separation into signal and noise subspaces and thereby retrieving the signal components alone. In the next section, application of SVD method of EEG extraction from contaminated EEG signals is explained by formulating the EEG enhancement problem in the SVD framework. Section 4 discusses the results on simulated as well as real signals and is followed by the conclusions.

2. SVD technique for signal separation

Singular value decomposition, a generalization of the eigenvalue–eigenvector representation of a square matrix, is one of the most important tools of numerical linear algebra. Singular value of a given data matrix contains information about the noise level, the energy, the rank of the matrix, etc. In fact, they provide the only computationally reliable technique for computing ranks of operators and selected basic objects of linear algebra [12, 8].

Suppose we have P observed signals $m_i(n)$ which are linear combinations of r source signals $s_i(n)$, corrupted by noise signals $n_i(n)$. This can be written as

$$m_i(n) = t_{i1}s_1(n) + \cdots + t_{ir}s_r(n) + n_i(n),$$
$$n = 1, 2, \ldots, q, \; i = 1, \ldots, P, \tag{1}$$

which can be written using matrix notation as

$$\boldsymbol{M} = \boldsymbol{TS} + \boldsymbol{N}, \tag{2}$$

where

$$\boldsymbol{M} = [\boldsymbol{m}(1), \boldsymbol{m}(2), \ldots, \boldsymbol{m}(q)] \in \mathbb{R}^{P\times q},$$

$$\boldsymbol{T} = [\boldsymbol{t}_1, \ldots, \boldsymbol{t}_r] \in \mathbb{R}^{P\times r},$$

$$\boldsymbol{S} = [\boldsymbol{s}(1), \boldsymbol{s}(2), \ldots, \boldsymbol{s}(q)] \in \mathbb{R}^{r\times q},$$

$$\boldsymbol{N} = [\boldsymbol{n}(1), \boldsymbol{n}(2), \ldots, \boldsymbol{n}(q)] \in \mathbb{R}^{P\times q},$$

$$\boldsymbol{m}(i) = [m_1(i), m_2(i), \ldots, m_P(i)]^{\mathrm{T}},$$

$$\boldsymbol{t}_i = [t_{1i}, t_{2i}, \ldots, t_{Pi}]^{\mathrm{T}},$$

$$\boldsymbol{s}(i) = [s_1(i), s_2(i), \ldots, s_r(i)]^{\mathrm{T}},$$

$$\boldsymbol{n}(i) = [n_1(i), n_2(i), \ldots, n_P(i)]^{\mathrm{T}},$$

with $m_j(i)$ $(j = 1, \ldots, P, i = 1, \ldots, q)$ being the ith sample of the signal recorded from the jth electrode, t_{ji} $(j = 1, \ldots, P, i = 1, \ldots, r)$ are the weighting coefficients, $s_j(i)$ $(j = 1, \ldots, r, i = 1, \ldots, q)$ is the ith sample of the jth source signal, and $n_j(i)$ $(j = 1, \ldots, P, i = 1, \ldots, q)$ is the ith sample of the noise present at the jth electrode.

The SVD of the matrix $\boldsymbol{M}$ can be written for $P < q$ as

$$\boldsymbol{M} = \boldsymbol{U\Sigma V}^{\mathrm{T}}, \tag{3}$$

where the matrices $U \in \mathbb{R}^{P \times P}$, $\Sigma \in \mathbb{R}^{P \times P}$, $V^{\mathrm{T}} \in \mathbb{R}^{P \times q}$ are such that

$$UU^{\mathrm{T}} = U^{\mathrm{T}}U = I_P, \qquad V^{\mathrm{T}}V = I_P,$$

and

$$\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \ldots, \sigma_p),$$

with

$$\sigma_1 > \sigma_2 > \cdots > \sigma_P > 0. \tag{4}$$

The elements σ_i of Σ are called the singular values of M and the columns of U and V are the called the singular vectors of M. The set of singular values are called the singular spectrum and the set is unique for a given M. These singular values give information about the noise level, signal energy, etc. The signal separation aspects of SVD techniques are explained well by Callaerts et al. [3].

If the signal present in the recordings is sufficiently strong compared to the noise, then the singular values of M can be partitioned into two groups as

$$\Sigma = \begin{bmatrix} \Sigma_{\mathrm{s}} & 0 \\ 0 & \Sigma_{\mathrm{n}} \end{bmatrix}, \tag{5}$$

where Σ_{s} contains r singular values associated with r source signals and Σ_{n} contains $P - r$ singular values associated with the noise. Corresponding to this, the matrix U can be partitioned as

$$U = [U_{\mathrm{s}} \quad U_{\mathrm{n}}], \tag{6}$$

where U_{s} contains the first r columns of U associated with the signal part and U_{n} contains the last $P - r$ columns of U associated with the noise part. Consequently, the span of the columns of U_{s} is called the signal subspace (of dimension r) and the span of the columns of U_{n} is called the noise subspace (of dimension $P - r$).

Since the signal and noise subspaces are orthogonal to each other, the solution to the signal separation problem can be presented as an orthonormal projection of the data onto the signal subspace, whereby noise contributions are eliminated. Formally, this can be written as

$$S = U^{\mathrm{T}}M = \begin{bmatrix} U_{\mathrm{s}}^{\mathrm{T}}M \\ U_{\mathrm{n}}^{\mathrm{T}}M \end{bmatrix} = \begin{bmatrix} S_{\mathrm{s}} \\ S_{\mathrm{n}} \end{bmatrix}, \tag{7}$$

where S_{s} is an $(r \times q)$ matrix containing estimates of the r source signals. It may be noted that the r rows of S_{s} are nothing but the first r columns of V (multiplied by the corresponding singular values) corresponding to the signal part.

Once the estimates of the source signals are found as $S_{\mathrm{s}} = U_{\mathrm{s}}^{\mathrm{T}}M$, one can construct a matrix F that contains only the source signal contribution in each measured signal as

$$F = U_{\mathrm{s}}S_{\mathrm{s}} = \sum_{i=1}^{r} \sigma_i u_i v_i^{\mathrm{T}}, \tag{8}$$

where u_i and v_i are the ith columns of U and V, respectively, and σ_i is the ith singular value of M. The signals in S_{s} are called the principal signals since each signal contribution in the recording can be found as a linear combination of these r principal signals.

3. Application to noise reduction in EEG signals

The first step in the formulation of the EOG artefact minimization problem is to obtain a suitable model for the measured EEG signals. The most commonly used model is the linear one because it is computationally simple and mathematically tractable. In this model, the measured signal is considered as the sum of pure EEG and a linear combination of EOG artefacts [14]. That is,

$$m_i(n) = t_{i1}s_1(n) + \cdots + t_{ir}s_r(n) + n_i(n),$$
$$i = 1, 2, \ldots, P, \tag{9}$$

where $m_i(n)$ is the signal measured from the ith electrode, $s_i(n)$, $i = 1, \ldots, r$, are the r EOG signals, and $n_i(n)$ is the pure EEG to be extracted.

It may be observed that the models described by Eqs. (1) and (9) are the same except that the desired signal to be extracted in Eq. (9) is $n_i(n)$ whereas it is $s_i(n)$ in Eq. (1). Hence, proceeding in similar lines as in Section 2, we get the matrix S (see Eq. (7)) as

$$S = U^{\mathrm{T}}M = \begin{bmatrix} U_{\mathrm{s}}^{\mathrm{T}}M \\ U_{\mathrm{n}}^{\mathrm{T}}M \end{bmatrix} = \begin{bmatrix} S_{\mathrm{s}} \\ S_{\mathrm{n}} \end{bmatrix}. \tag{10}$$

Since $n_i(n)$ is the desired signal in this application (see Eq. (9)), the $(P - r) \times q$ matrix S_{n} contains the

estimates of the $P - r$ desired signals (EEG). Hence, the matrix $\boldsymbol{F}$ (of size $P \times q$) that contains only the EEG contribution in each measured signal can be given as

$$\boldsymbol{F} = \boldsymbol{U}_n \boldsymbol{S}_n = \sum_{i=r+1}^{P} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T. \tag{11}$$

Thus, the rows of $\boldsymbol{S}_n$ give the artefact minimized EEG signals.

4. Results and discussion

We divide this section into two subsections in which the first subsection discusses the results obtained from the studies performed with known signals. The second subsection discusses the results obtained from the studies on recorded EEG signals.

4.1. Results on simulated signals

In order to get an idea about the effectiveness of SVD techniques in minimizing EOG artefacts from contaminated EEG signals, we performed studies with known signals. The simulated EEG signal [10] sampled at 100 Hz has been used as the desired signal. This simulated EEG signal is corrupted by linearly combining EOG signals recorded from left and right eye positions. The signal to noise ratio (SNR) of this EOG contaminated EEG signal was -9.5 dB. Studies were performed with one and two EOG signal channels and compared. The matrix $\boldsymbol{M}$ (2×400 for one EOG signal scheme and 3×400 for two EOG signal schemes) was constructed with these signals. The SVD of these $\boldsymbol{M}$ matrices was computed. Then, the matrix $\boldsymbol{S} = \boldsymbol{U}^T\boldsymbol{M}$ was computed. In order to correctly identify which rows of $\boldsymbol{S}$ correspond to the artefact minimized EEG signals, we need to approximately group the singular values of $\boldsymbol{M}$. To do this groupings, we require to consider the following points: (i) the average power of the EOG signal is much larger than that of EEG signals, and (ii) the EOGs from left and right eye positions are highly correlated. As a consequence of (ii), the effective number of linearly independent EOG signals is one, i.e., $r = 1$. As a result of this (i.e. $r = 1$) and fact (i), the highest singular value corresponds to the EOG signal and the second singular value corresponds to EEG signal. Hence, the 2nd row of $\boldsymbol{S}$ gives the EOG minimized EEG signal. The time plots are shown in Fig. 1. Fig. 1(a) shows four

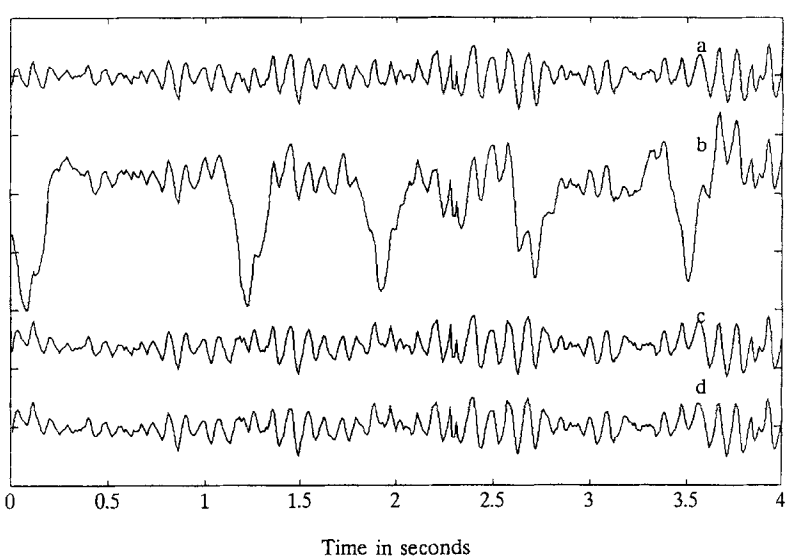

Fig. 1. Time plots of the signal before and after processing (simulated case): (a) simulated EEG signal, (b) EOG contaminated EEG signal, (c) EOG minimized EEG signal with one reference input and (d) with two reference inputs.

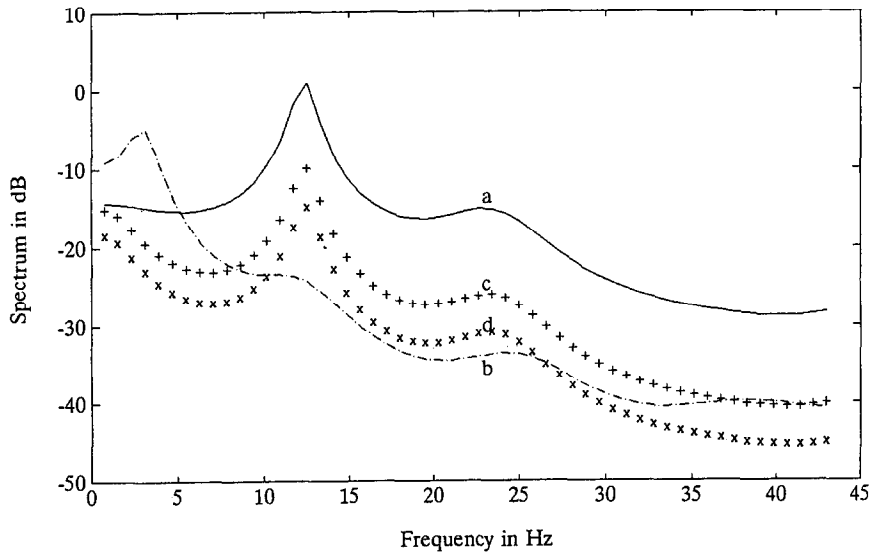

Fig. 2. LP spectra of the signals corresponding to Fig. 1.

seconds of simulated EEG signal which is the desired signal. Fig. 1(b) is the EOG contaminated EEG signal (by linearly combining EOGs from left and right eye positions to the desired EEG signal) at -9.5 dB input SNR. Figs. 1(c) and (d) are the time plots of the EOG minimized EEG signals using the SVD technique with one and two EOG signals, respectively. Comparing these two time plots, both look almost similar and compare well with the desired signal. This is also evident from the output SNR values of the signals shown in Figs. 1(c) and (d) which are 9.95 and 10.07 dB, respectively. These results imply that the use of two EOG signals does not yield significant additional performance gain compared to the scheme with only one EOG signal.

Fig. 2 shows the linear prediction (LP) spectra [9] of the signals shown in Fig. 1. Fig. 2(b) is the LP spectrum of the EOG contaminated EEG signal which gives an additional low frequency peak as compared to Fig. 2(a) which is the LP spectrum of the desired EEG signal. Figs. 2(c) and (d) show the LP spectra of EOG minimized EEG signals obtained from the SVD technique with one and two EOG signals schemes, respectively. It may be observed that both the schemes give almost identical LP spectra and they match well with the LP spectrum of the desired EEG signal. This is clear from the time plots also. This study shows the effectiveness of SVD technique in minimizing EOG artefacts from contaminated EEG signals.

4.2. Results on recorded signals

Similar studies were performed with recorded EEG signals also. We considered the EOG contaminated EEG signal recorded from Fp1 position on the scalp as one input signal. EOG signals from left and right eye positions were considered as two other inputs. Studies were performed with one and two EOG inputs and the results reveal the same observations as we have observed in the previous section. The time plots and the LP spectra are shown in Figs. 3 and 4, respectively.

The SVD techniques was applied to a set of four recordings, which is shown in Fig. 5. The first two signals (Figs. 5(a) and (b)) were recorded from Fp1 and Fp2 electrode positions on the scalp using a Nihon-Kohden EEG machine. In these signals, the low frequency high amplitude activities are due to eye movements (EOG). The other two signals

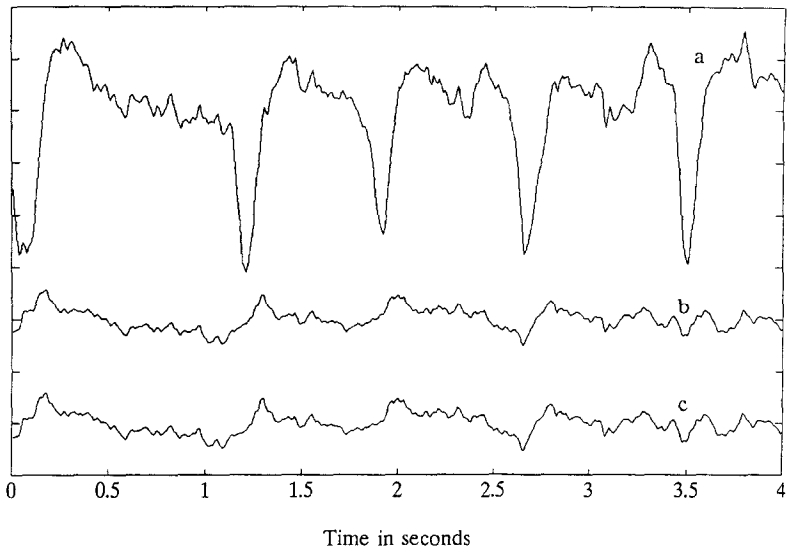

Fig. 3. Time plots of the signals before and after processing (recorded case): (a) EEG signal recorded from Fp1 position, (b) EOG minimized EEG signal with one reference input and (c) with two reference inputs.

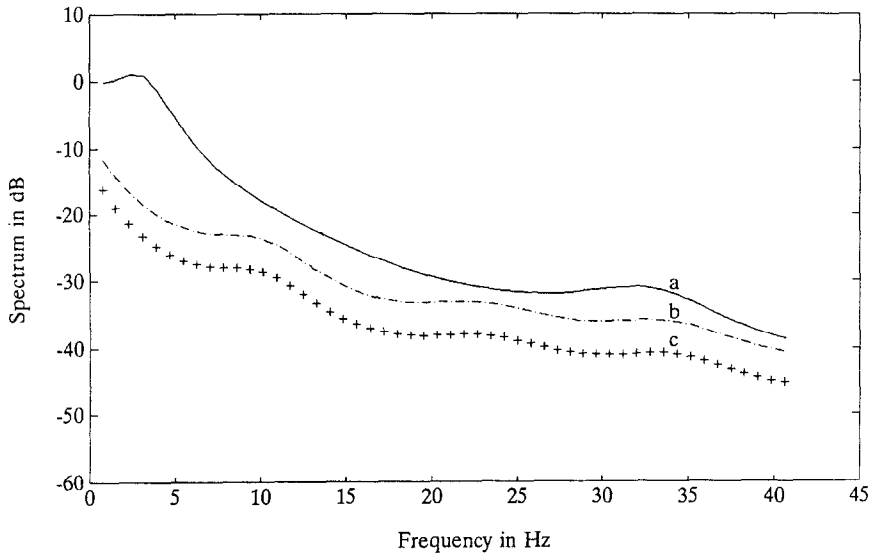

Fig. 4. LP spectra of the signals corresponding to Fig. 3.

(Figs. 5(c) and (d)) were recorded from left and right eye positions, respectively, using the same machine simultaneously with other recordings. These figures (5(c) and (d)) show the eye movement signals that are reflected in the first two recordings of EEG signals. Four seconds of data was used in the simulation.

The matrix $\boldsymbol{M}$ (4×400) was constructed with these four channels of the recorded data. Then, the SVD of $\boldsymbol{M}$ was computed and then the matrix

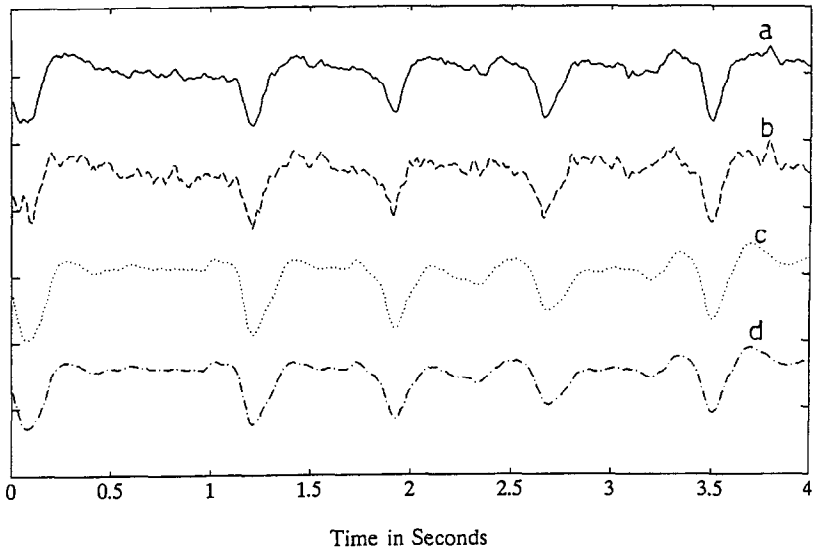

Fig. 5. Time plots of the signals recorded: (a) EOG contaminated EEG signal recorded from Fp1 position on the scalp, (b) EOG contaminated EEG signal recorded from Fp2 position on the scalp, (c) EOG signal recorded from left eye position and (d) EOG signal recorded from right eye position.

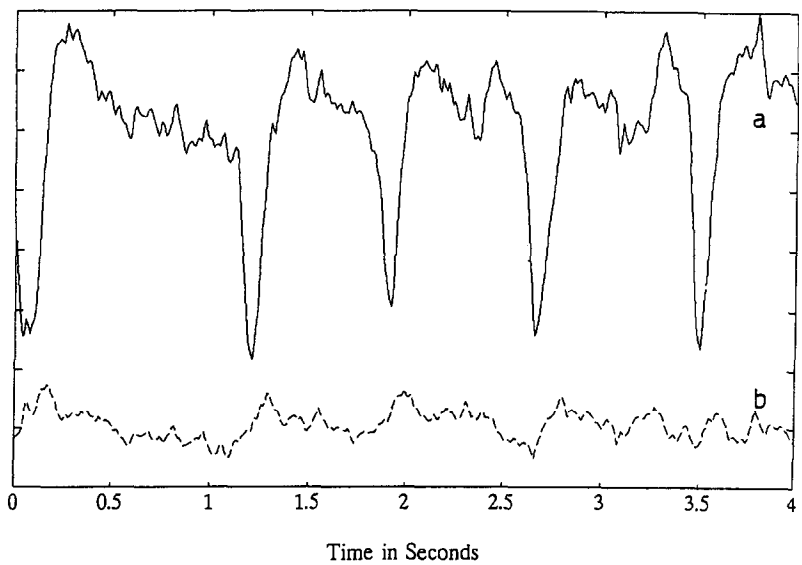

Fig. 6. Signals before and after processing in the time domain (recorded case): (a) unprocessed EOG contaminated EEG recorded from Fp1 position and (b) extracted EEG signal.

$\boldsymbol{S} = \boldsymbol{U}^{\mathrm{T}}\boldsymbol{M}$ (see Eq. (10)) was computed. Since we have two EEG signals here, the second and third singular values correspond to the EEG signals. Subsequently, the second and third rows of $\boldsymbol{S}$ contain the artefact free EEG signals. These results are shown in Figs. 6 and 7. Fig. 6(a) is the recorded EEG signal (at Fp1 position) which contains high EOG activities, whereas in Fig. 6(b) these EOG

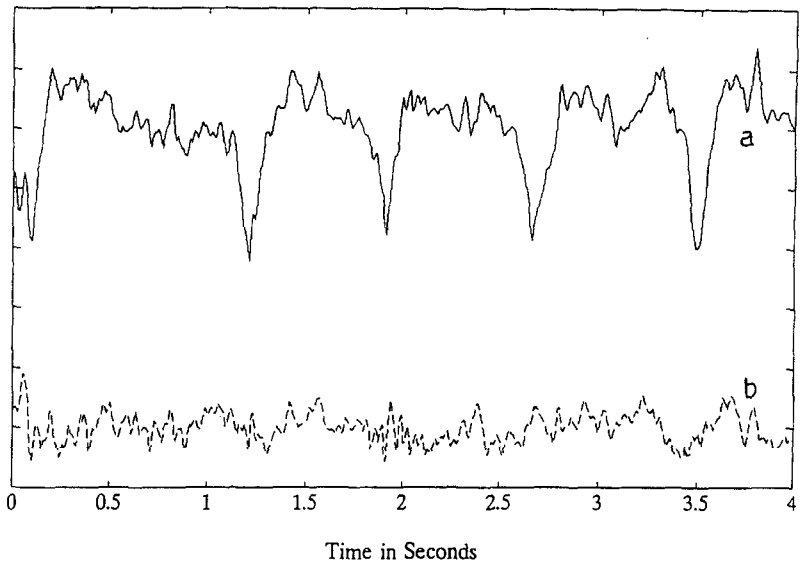

Fig. 7. Signals before and after processing in time domain (recorded case): (a) unprocessed EOG contaminated EEG recorded from Fp2 position and (b) extracted EEG signal.

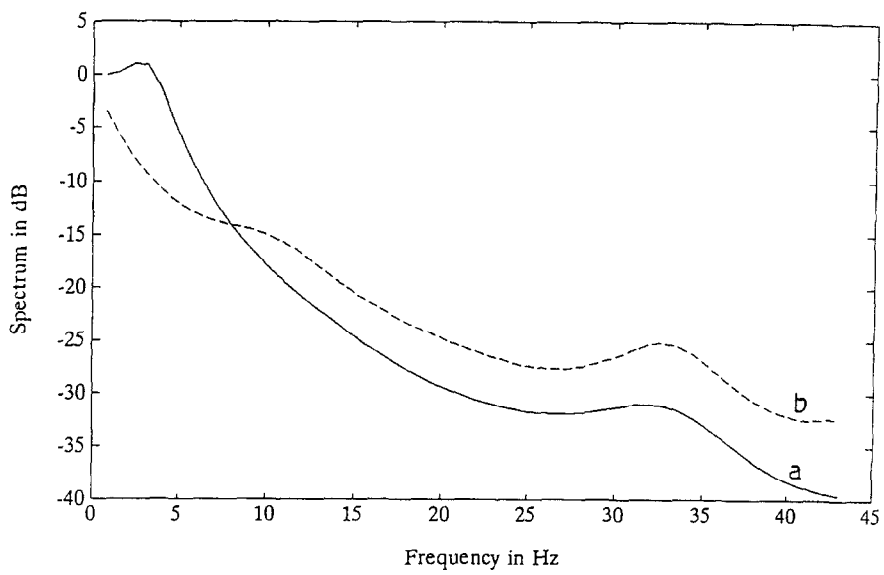

Fig. 8. Smoothed LP spectra of signals shown in Fig. 6.

activities have been effectively minimized. The corresponding signals for Fp2 electrode position is shown in Fig. 7. Here also, we observe that the EOG activities have been effectively minimized.

The smoothed LP spectra of the recorded and processed signals were also computed for evaluating the extent of artefact minimization. Fig. 8(a) is the LP spectrum of the EOG contaminated EEG

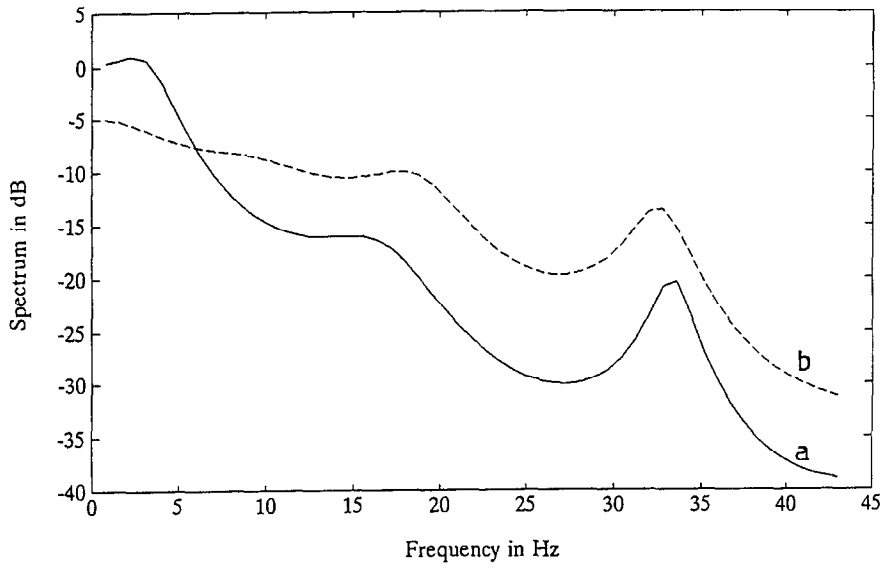

Fig. 9. Smoothed LP spectra of signals shown in Fig. 7.

signal recorded from Fp1 electrode position. This shows a peak at low frequency which indicates the presence of EOG artefacts as seen from the time plots. However, in Fig. 8(b), this peak has been effectively removed whereas other peaks remain unaltered. This shows the effectiveness of the proposed algorithm in minimizing the EOG artefacts. Fig. 9 shows the corresponding spectra for the signals shown in Fig. 7. Similar observations as in Fig. 8 can be made from Fig. 9.

From the above observations we conclude that the SVD is a very effective tool for minimizing EOG artefacts from contaminated EEG signals.

5. Conclusions

In this paper, we explored the application of singular value decomposition for extracting actual EEG signals from EOG contaminated EEG signals. The key point which made the SVD technique suitable for this EEG enhancement problem is the orthogonality between the EEG and EOG signals. The problem has been modeled in such a way that it can be directly fit into the SVD framework. The proposed scheme has been evaluated with simulated signals using SNR, time plots and the LP spectra as the measures. In these studies, it was observed that the scheme with one EOG channel works as good as the scheme with two EOG channels. This method has also been applied to a set of four channels of data recorded from left and right eye positions and Fp1 and Fp2 electrode positions on the scalp. Studies on this data set show that the SVD based technique is very effective in minimizing EOG artefacts from noisy EEG signals. Smoothed power spectra and time plots were used to substantiate the above observations. From all these observations, we conclude that SVD based technique is very powerful in extracting actual EEG signals in the presence of EOG artefacts.

References

[1] J.S. Barlow, "Computerized electroencephalography in perspective", *IEEE Trans. BME*, Vol. 26, 1979, pp. 377–391.

[2] J.S. Barlow, "Artefact processing (rejection and minimization) in EEG data processing", in: F.H. Lopes de Silva, W. Storm van Leeuwen and A. Remond, eds., *Handbook of Electroencephalography and Clinical Neurophysiology* (Revised Series), Vol. 2, Elsevier, Amsterdam, 1986, pp. 1–62.

[3] D. Callaerts et al., "On-line algorithm for signal separation based on SVD", in: E.F. Deprettere, ed., *SVD and Signal Processing: Algorithms, Applications and Architectures*, Elsevier, Amsterdam, 1988, pp. 269–276.

[4] A.S. Gevins, C.L. Yeager, G.M. Zeitlin, S. Ancoli and M.F. Dedon, "On-line computer rejection of EEG artefact", *Electroenceph. Clin. Neurophysiol.*, Vol. 42, 1977, pp. 267–274.

[5] G. Gratton, M.G.H. Coles and E. Donchin, "A new method for off-line removal of ocular artefacts", *Electroenceph. Clin. Neurophysiol.*, Vol. 55, 1983, pp. 468–484.

[6] H.H. Jasper, "Report of the committee on methods of clinical examination in electroencephalography", *Electroenceph. Clin. Neurophysiol.*, Vol. 10, 1958, pp. 370–375.

[7] B.W. Jervis, E.C. Ifeachor and E.M. Allen, "The removal of ocular artefacts from electroencephalogram: a review", *Med. Biol. Eng. Comput.*, Vol. 26, 1988, pp. 2–12.

[8] V.C. Klema and A.J. Laub, "The singular value decomposition: its computation and some applications", *IEEE Trans. Autom. Control*, Vol. 25, 1980, pp. 164–176.

[9] J. Makhoul, "Linear prediction: A tutorial review", *Proc. IEEE*, Vol. 63, 1975, pp. 561–580.

[10] S.V. Narasimhan and D. Narayana Dutt, "Software simulation of the EEG", *J. Biomed. Eng.*, Vol. 7, 1985, pp. 275–281.

[11] D. Papakostopoulos, A. Winter and P. Newton, "New technique for the control of eye potential artefacts in multichannel CNV recordings", *Electroenceph. Clin. Neurophysiol.*, Vol. 34, 1973, pp. 651–653.

[12] D.W. Tufts and R. Kumaresan, "Singular value decomposition and improved frequency estimation using linear prediction", *IEEE Trans. Acoust. Speech Signal Process.*, Vol. 30, 1982, pp. 671–675.

[13] M.M.C. Van Den Berg-Lenssen et al., "Correction of ocular artefacts in EEGs using an autoregressive model to describe the EEG: a pilot study", *Electroenceph. Clin. Neurophysiol.*, Vol. 73, 1989, pp. 72–83.

[14] M.M.C. Van Den Berg-Lenssen et al., "Comparison of two methods for correcting ocular artefacts in EEGs", *Med. Biol. Eng. Comput.*, Vol. 32, 1994, pp. 501–511.

[15] R. Verleger et al., "Correction of EOG artefacts in event related potentials of the EEG: aspects of reliability and validity", *Psychophysiology*, Vol. 19, 1982, pp. 472–480.